



Kubernetes + Google

5 lessons learned from 8+ years of containers

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Kubernetes

κυβερνήτης: *Greek for “pilot” or “helmsman of a ship”*
the open source cluster workload manager from Google

- Manage applications, not machines
- Inspired by Google’s internal systems
  - >10 years experience running containerized apps
  - >2B containers / week
- Extensible & portable
- Could run anywhere
- Apache 2.0 licensed
- Written in Go



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Cluster

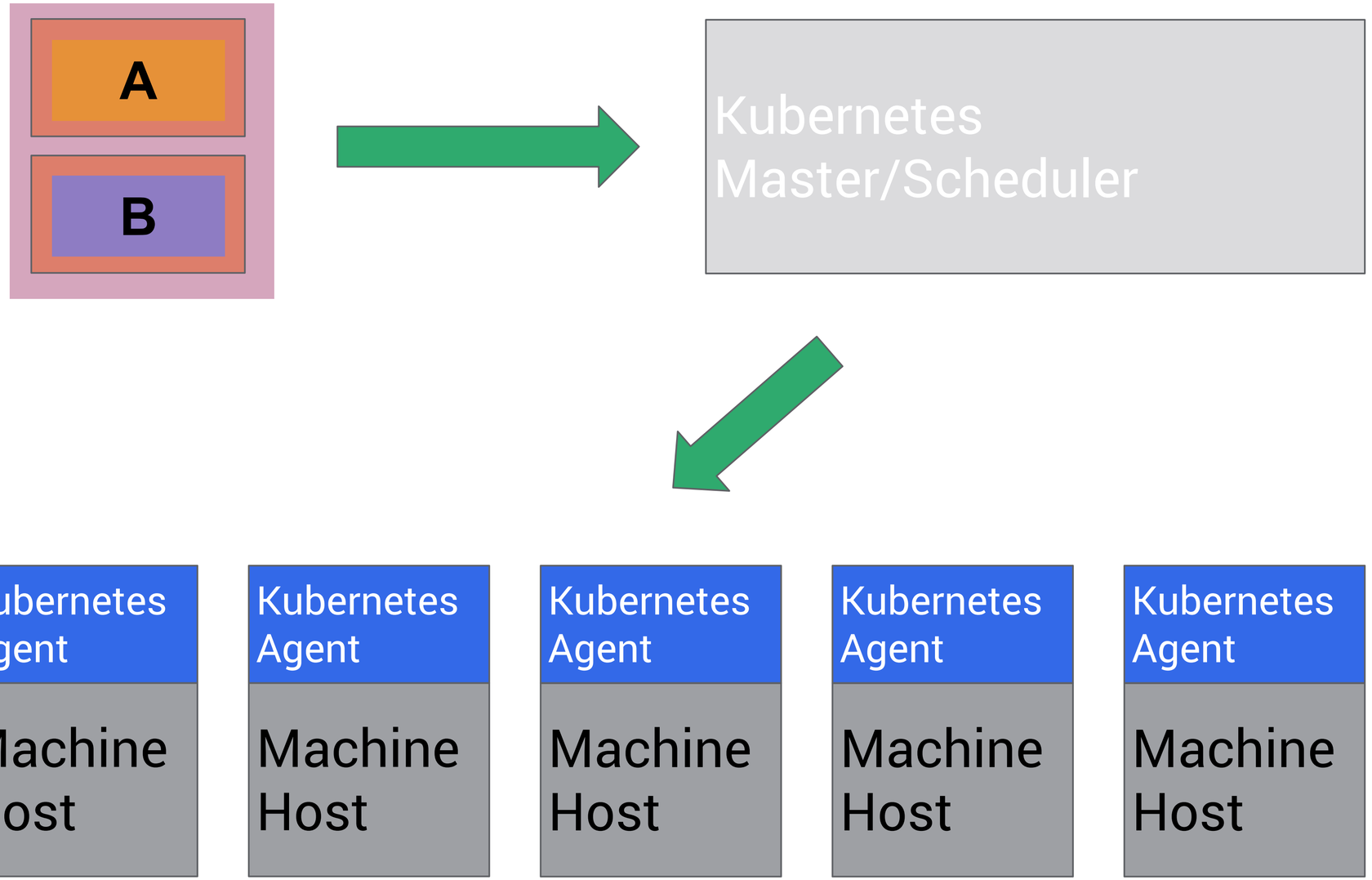


Kubernetes
Master/Scheduler

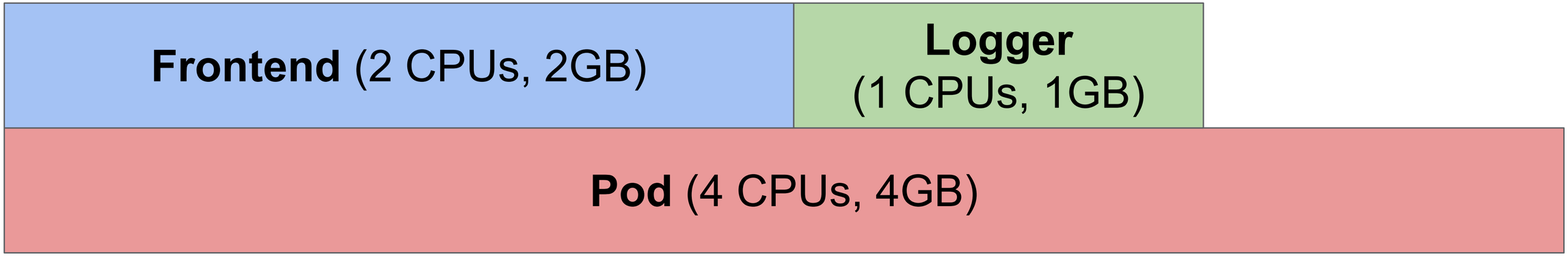
Kubernetes Agent	Kubernetes Agent	Kubernetes Agent	Kubernetes Agent	Kubernetes Agent	Kubernetes Agent	Kubernetes Agent
Machine Host	Machine Host	Machine Host	Machine Host	Machine Host	Machine Host	Machine Host

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Lesson 1: Pods



Lesson 1: Pods

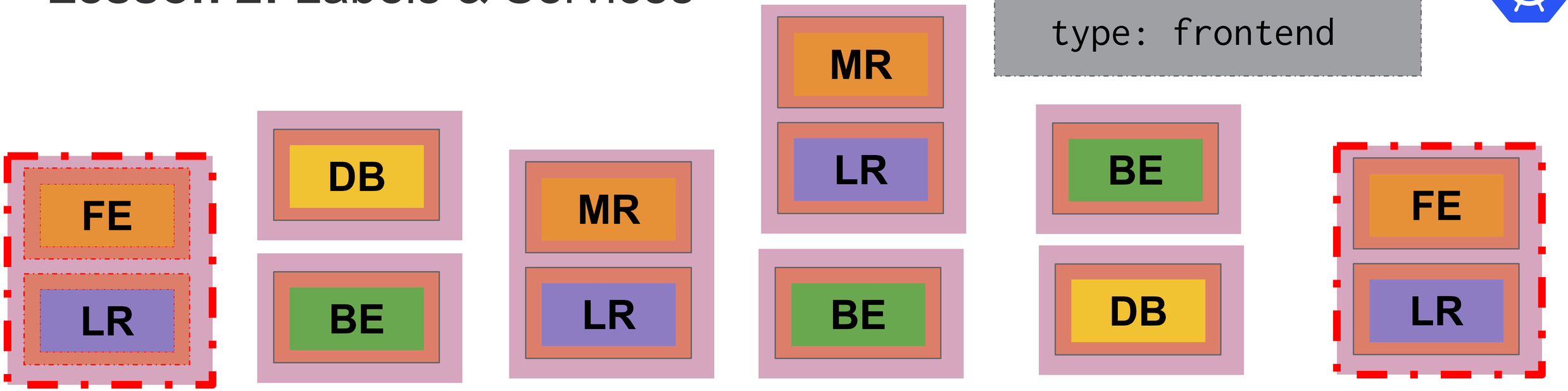


- Easy (and thorough) usage accounting and limiting
- Enables resource sharing

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Lesson 2: Labels & Services

```
labels:
  type: frontend
```

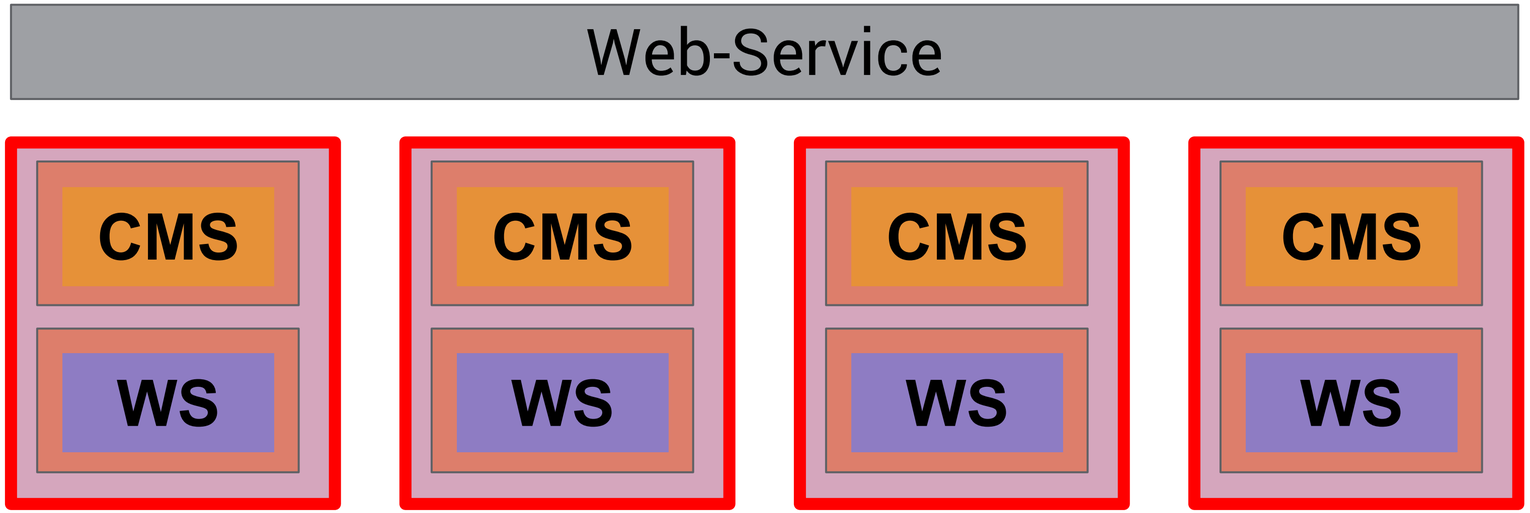


Kubernetes - Master/Scheduler

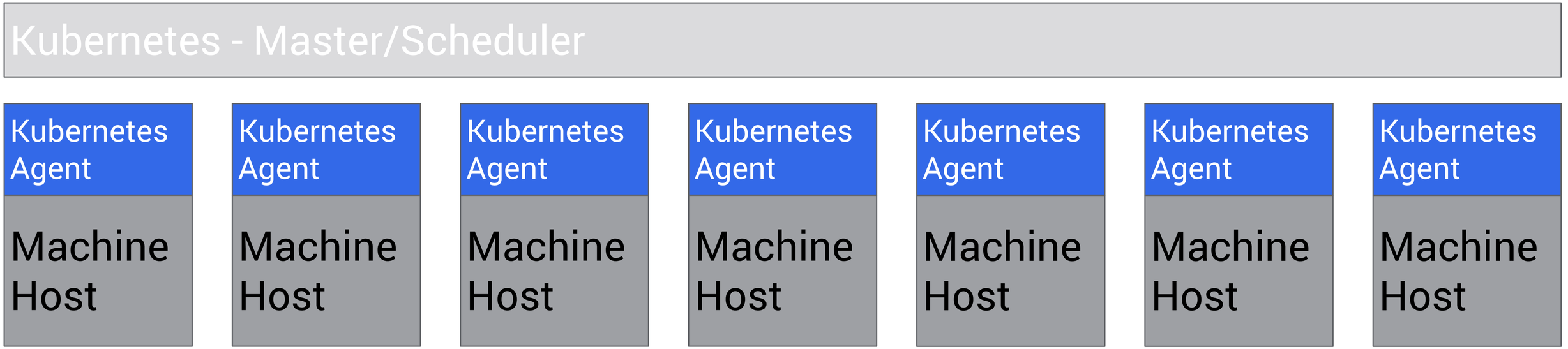
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Google

Lesson 2: Labels & Services

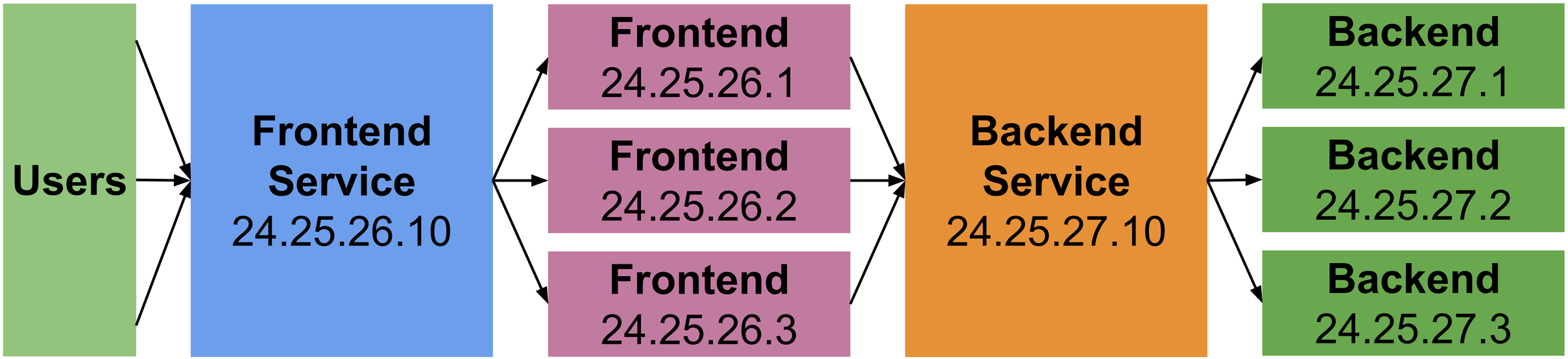



```
id: web-service
port: 9000
selector:
  tier: frontend
  env: production
```



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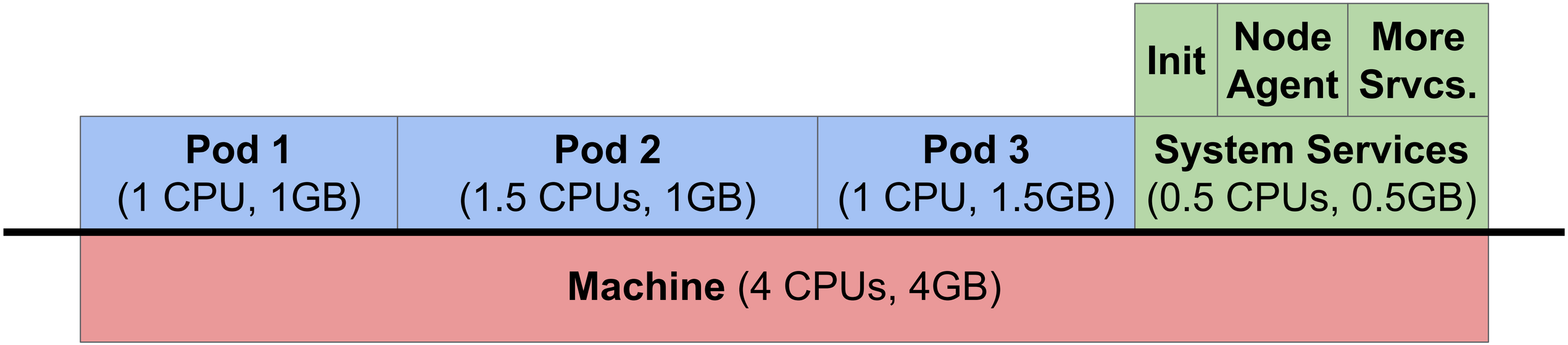
Lesson 3: IP per Pod and IP per Service



- Makes addressing any entity easy (no need for host, port location tuple)
- Users should never share a namespace they don't control
- Port brokering and scheduling is very difficult

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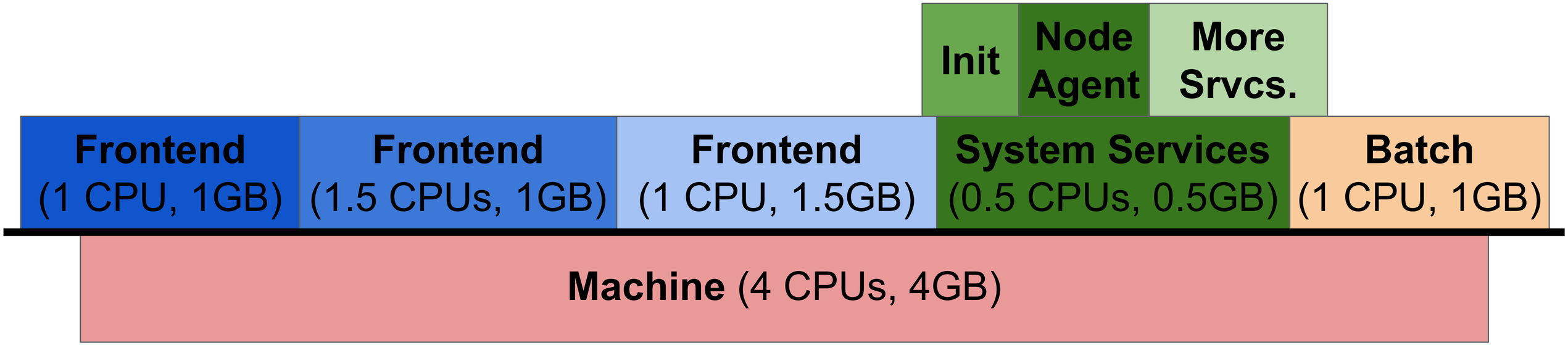
Lesson 4: *Everything* is in a Container



- If something is not containerized, nothing is
- Everything has a limit
- Detailed accountability of resource usage
- We use subcontainers extensively for accounting

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Lesson 5: Resource Tiers & Priority



- Enables selling “tiers” of resources with varying guarantees
- Increases utilization
- Allows handling out of resource conditions (what pod do I kill?)
- Must handle out of resource conditions well
- Intent-based at higher layers

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Questions?



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Kubernetes
github.com/googlecloudplatform/kubernetes

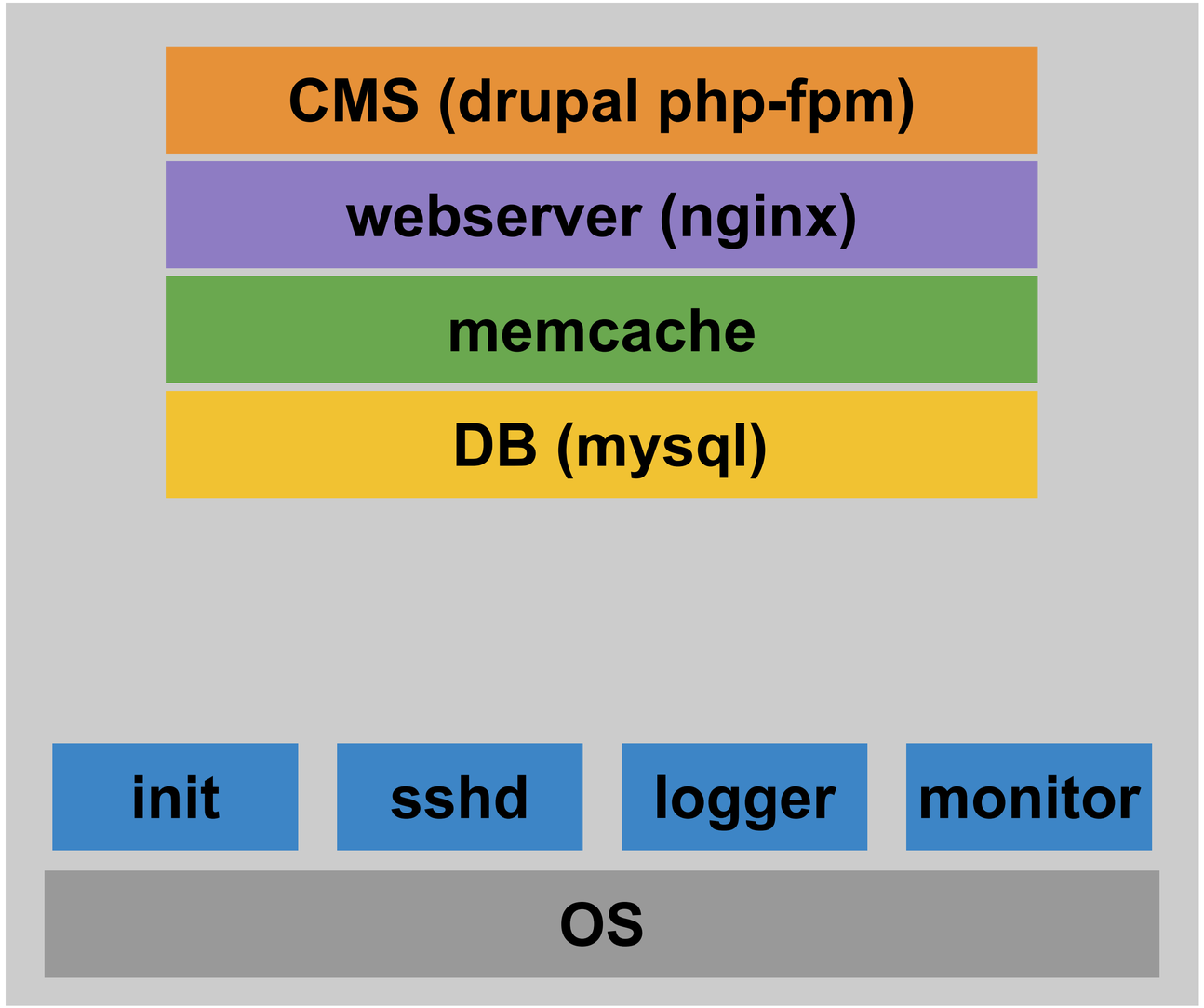
Mailing List: google-containers@googlegroups.com
IRC: #google-containers on Freenode

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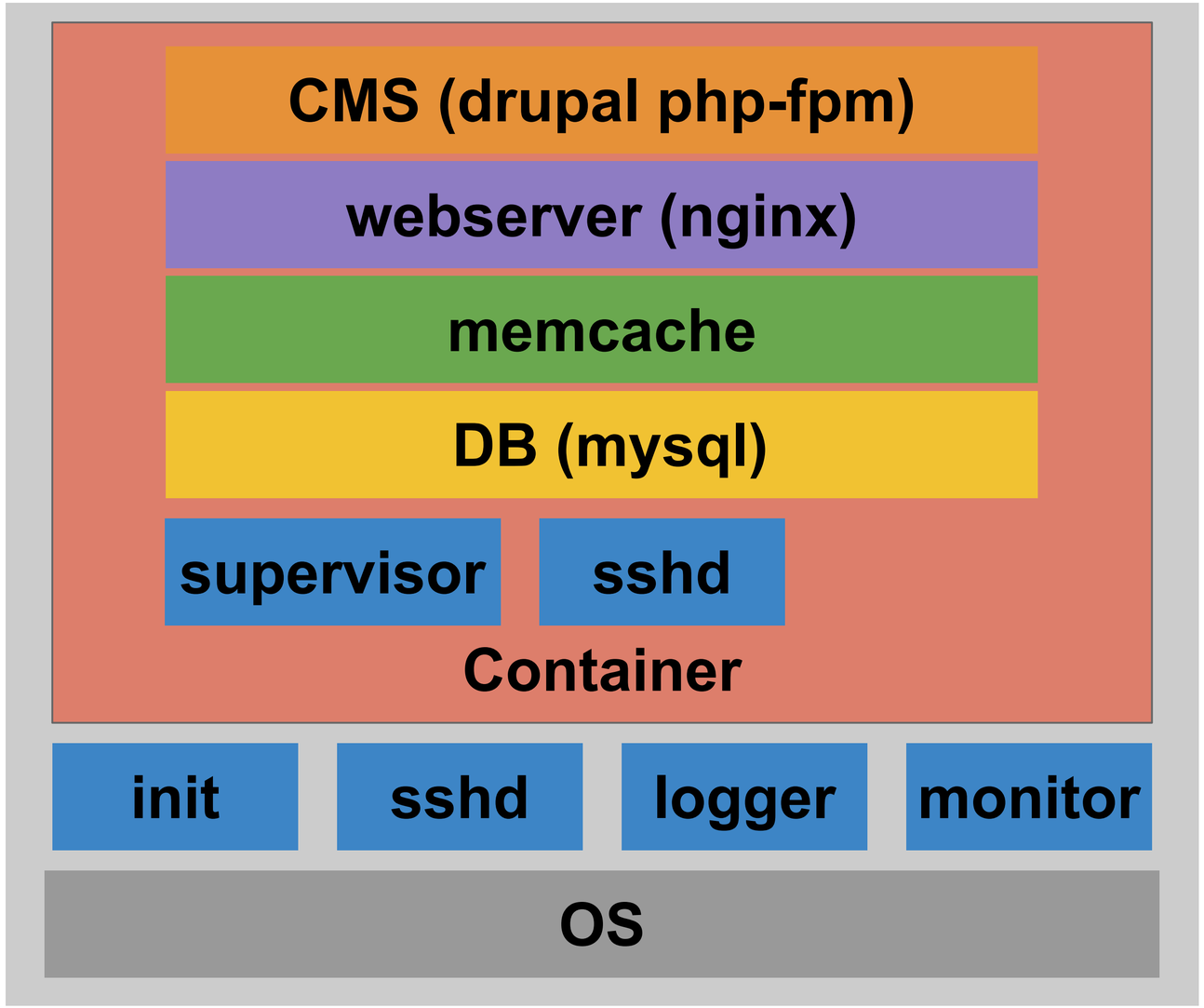
Backup



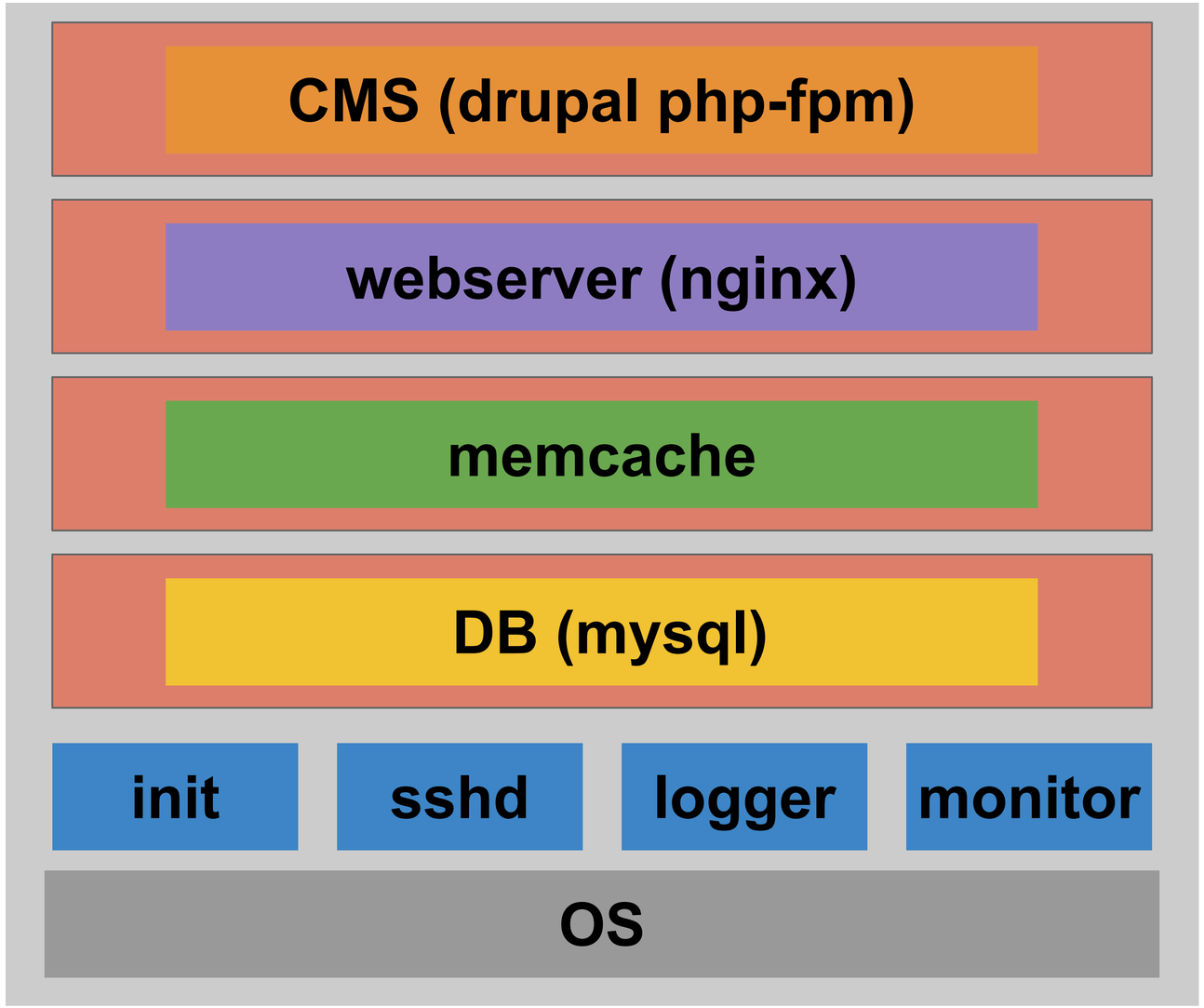
Application stack on single host



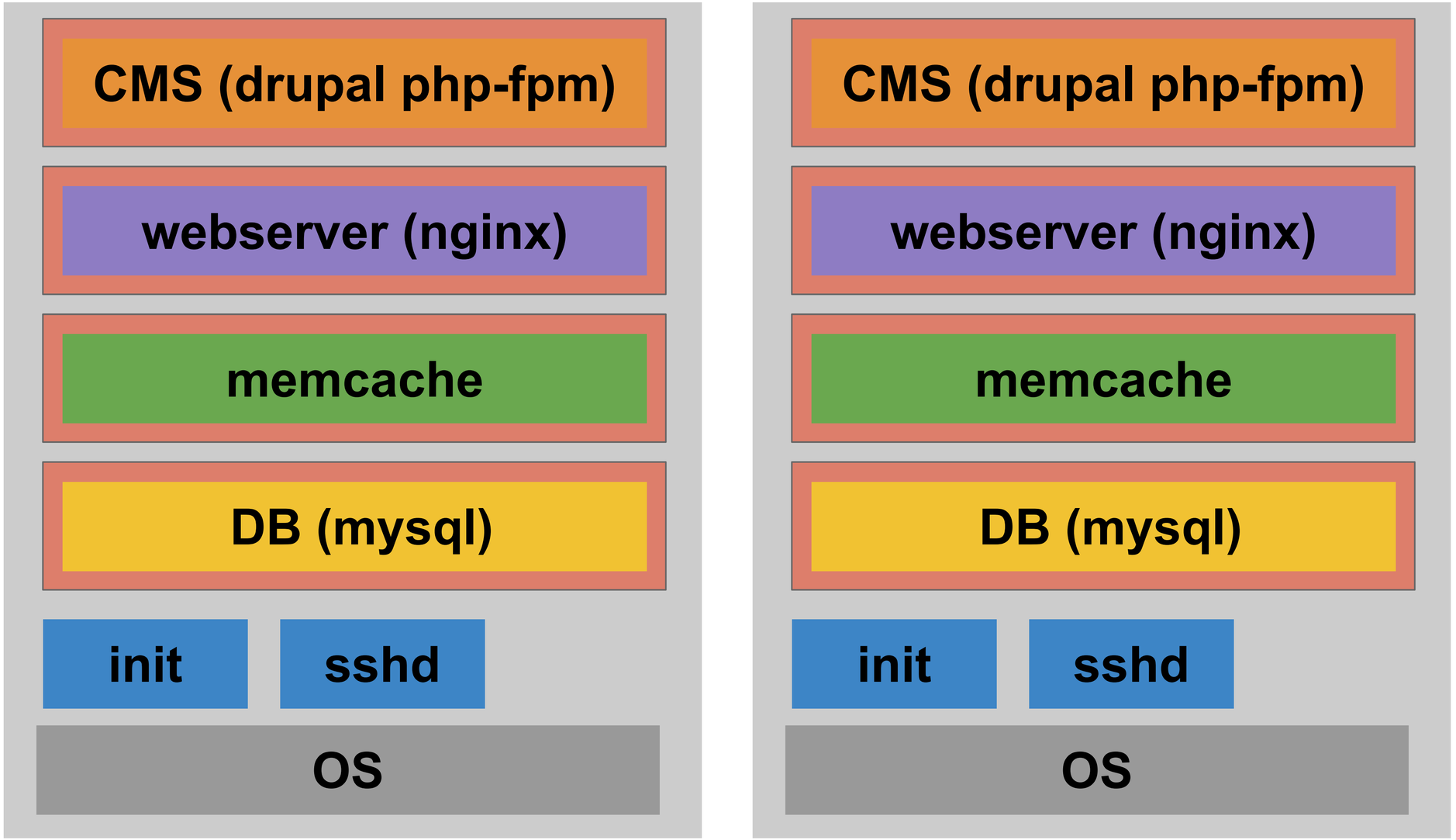
Application stack in single container



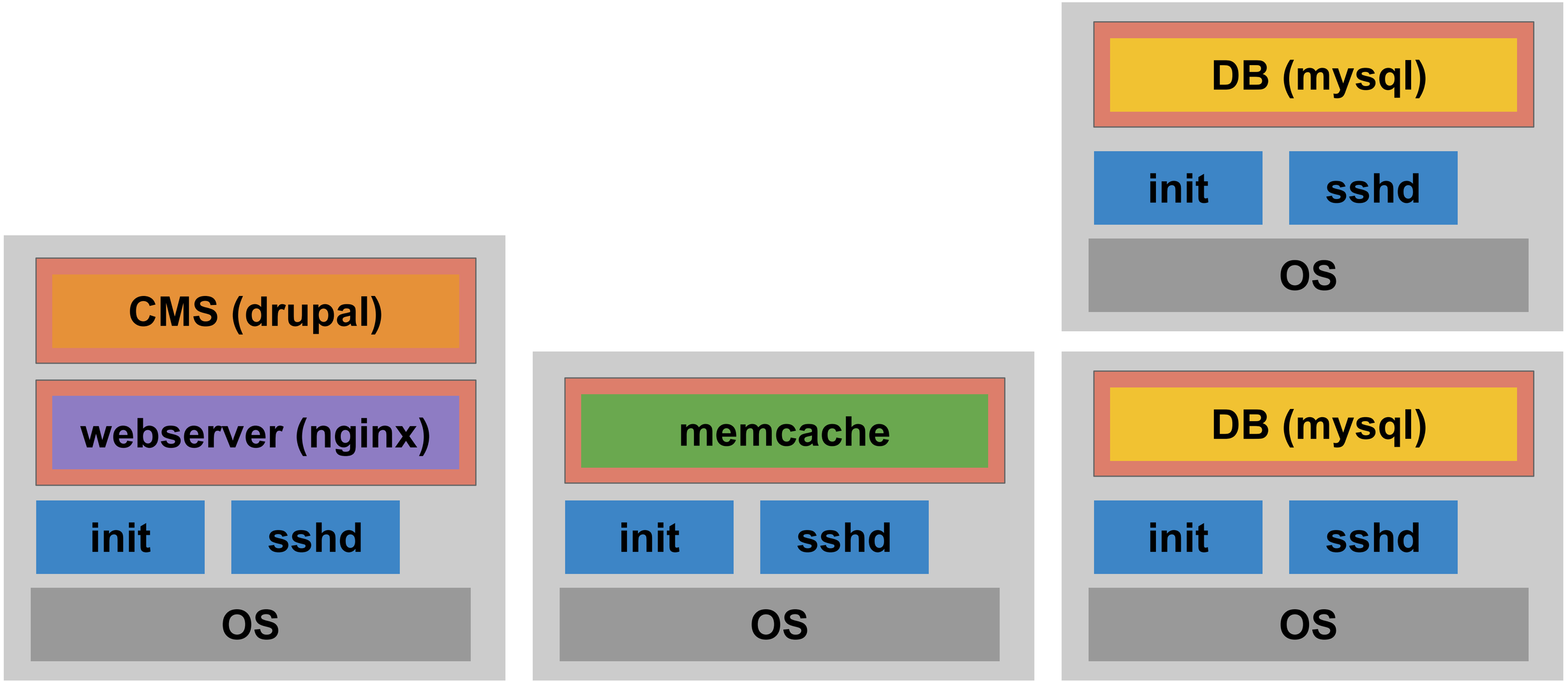
Application stack in separate containers



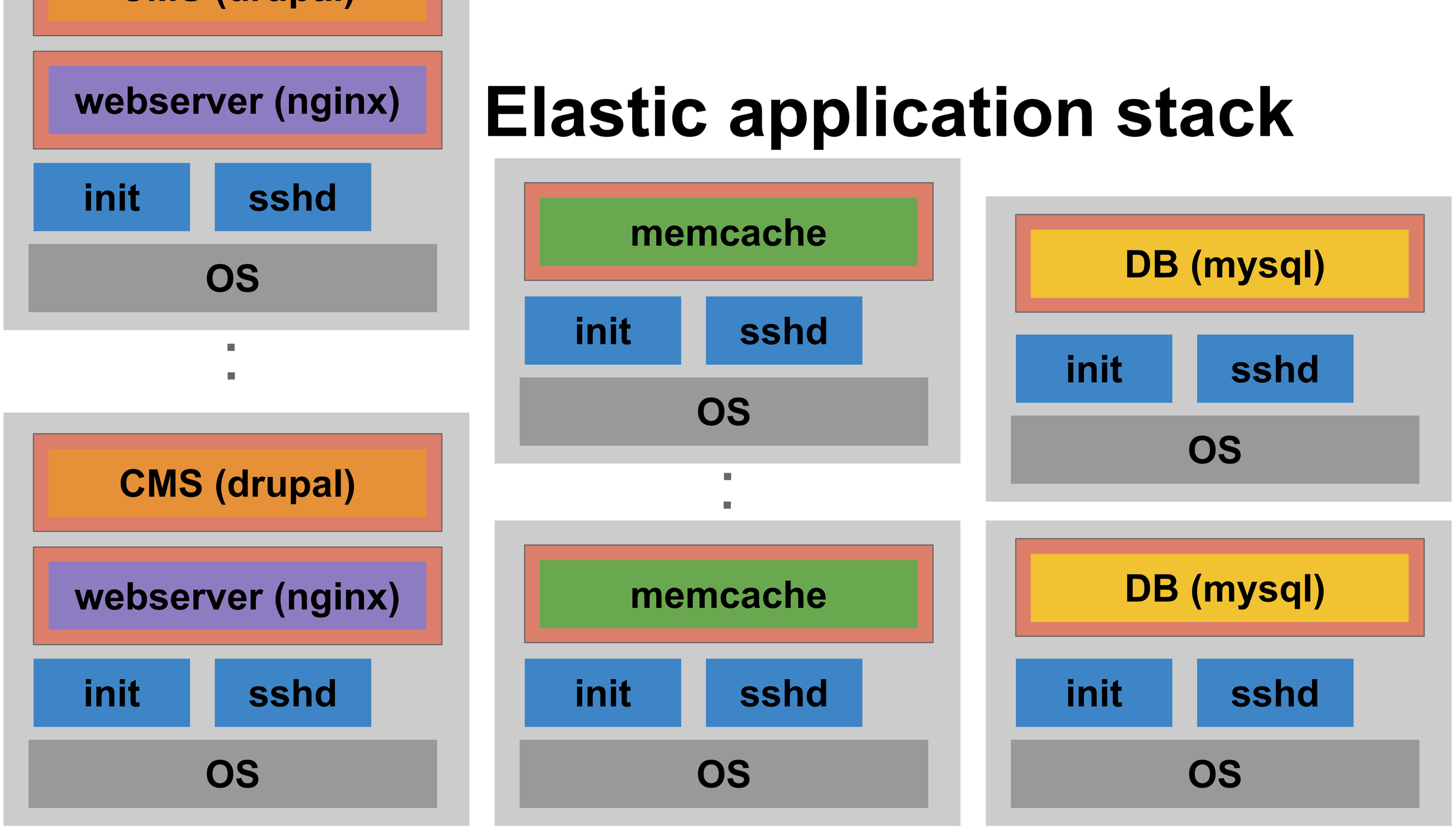
Replicated application stack



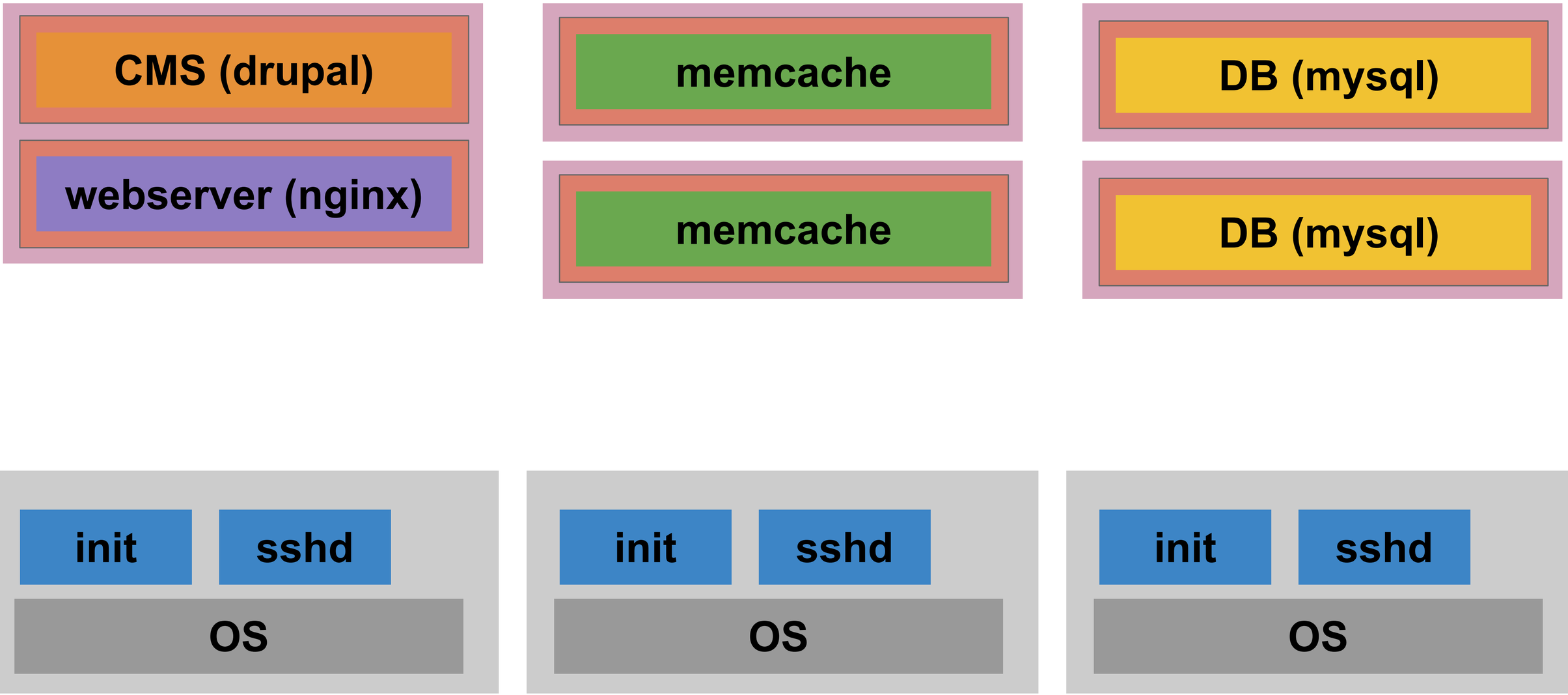
Distributed application stack

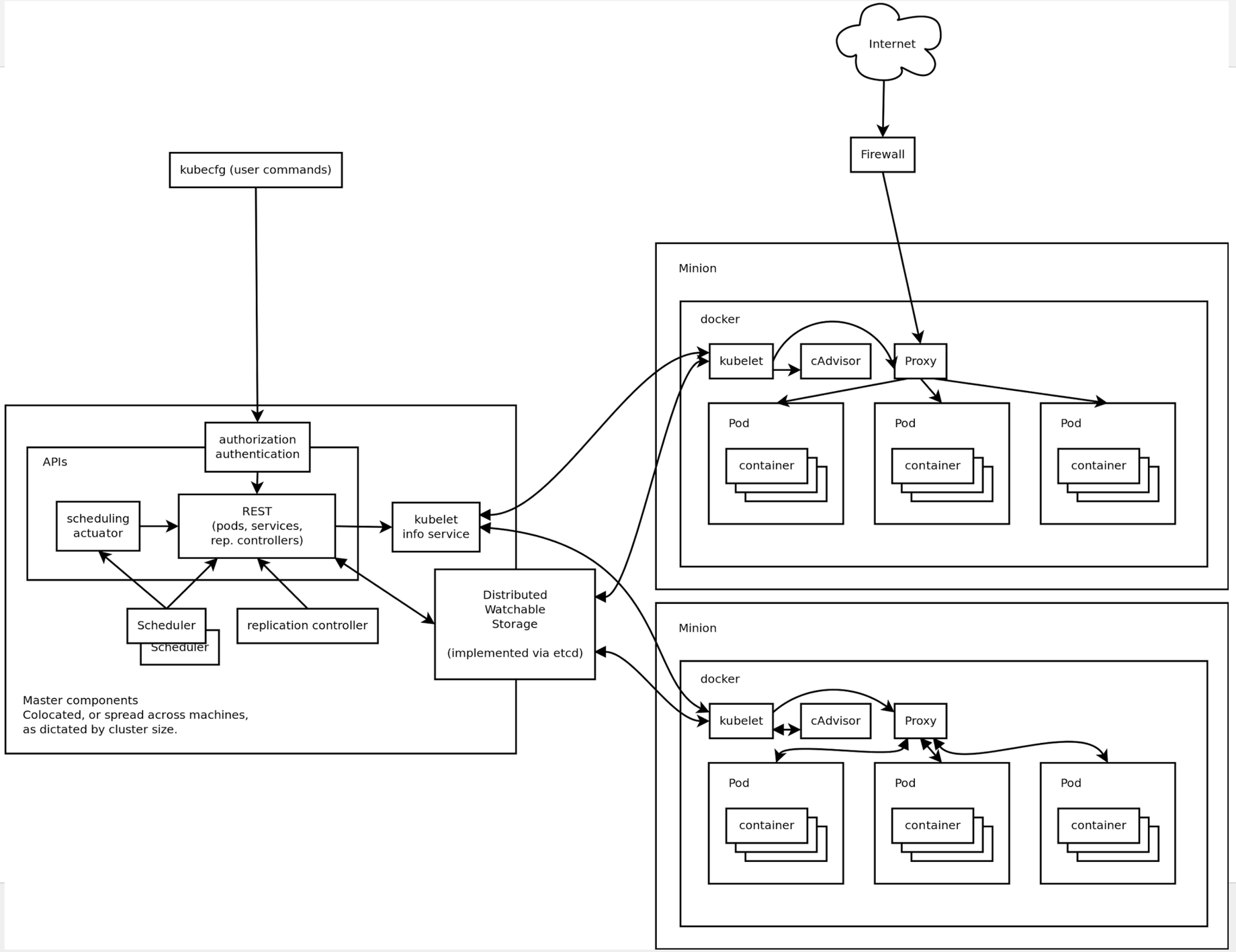


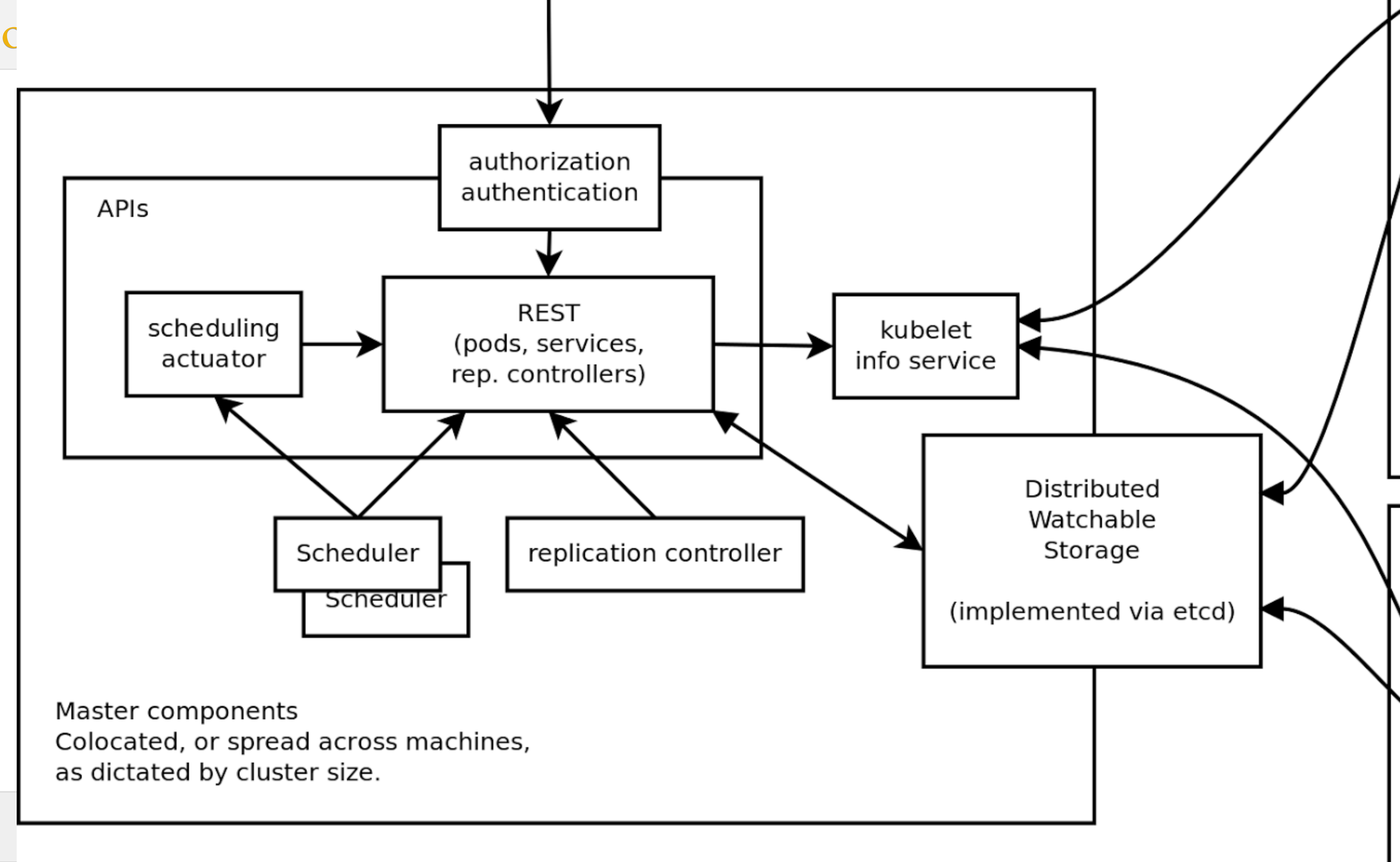
Elastic application stack

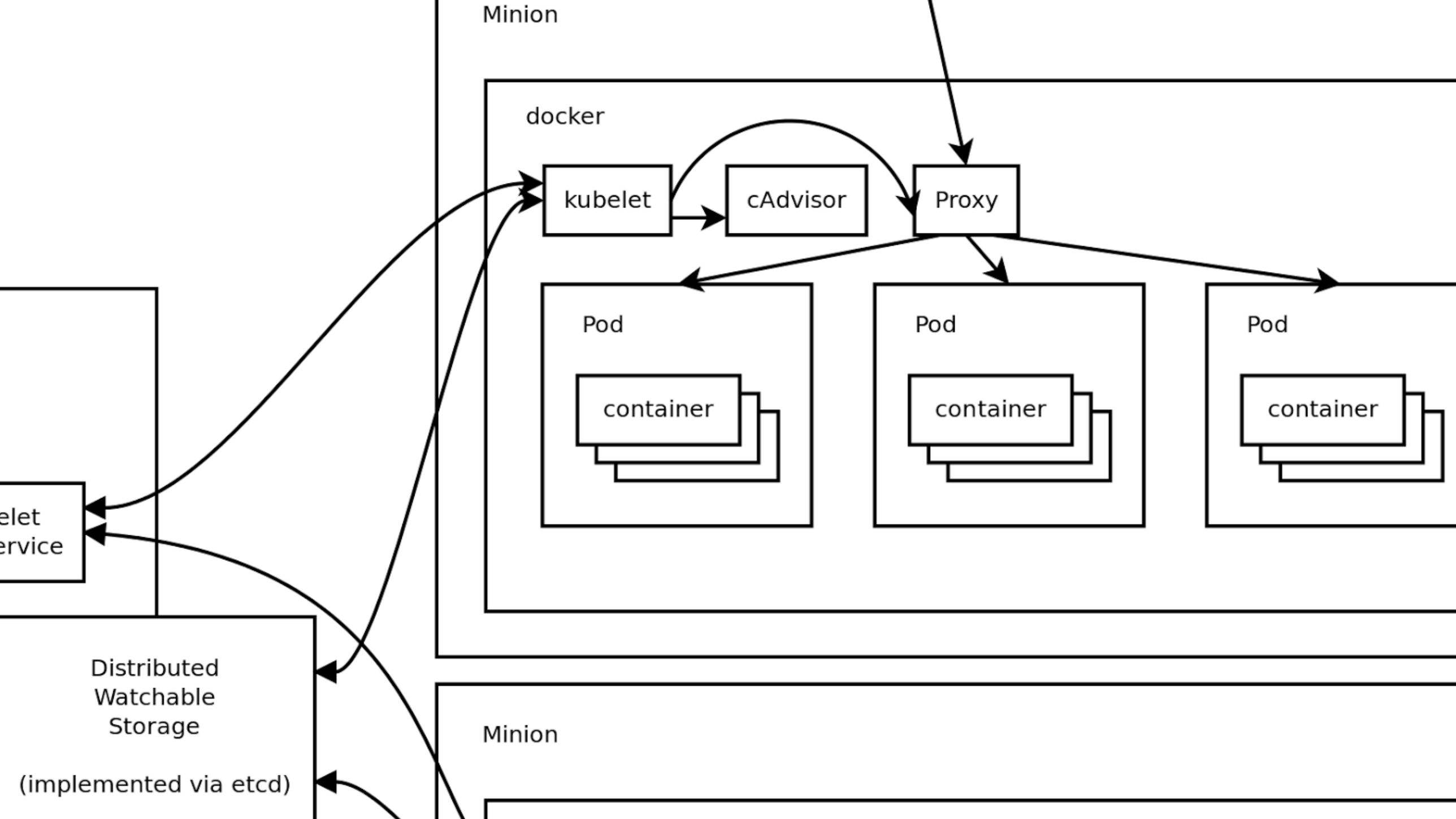


Liberated application stack




Internet
Firewall
kubecfg (user commands)
Minion
docker
kubelet
cAdvisor
Proxy
Pod
container
Pod
container
Pod
container
authorization
authentication
APIs
scheduling
actuator
REST
(pods, services,
rep. controllers)
kubelet
info service
Distributed
Watchable
Storage
(implemented via etcd)
Scheduler
Scheduler
replication controller
Master components
Colocated, or spread across machines,
as dictated by cluster size.
Minion
docker
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Minion
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